



Can Swarm Intelligence Make AI Collectives Predictable?

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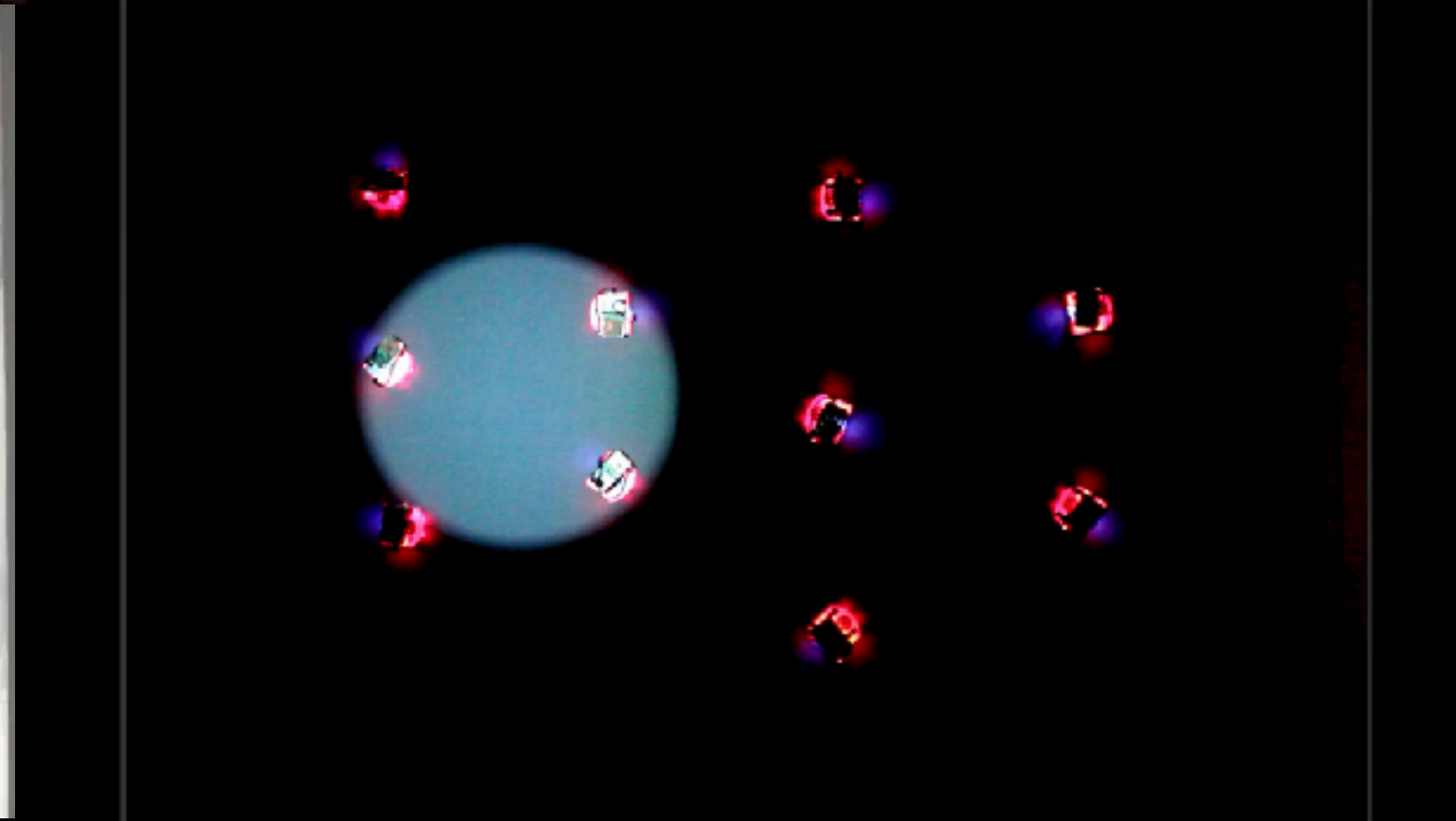
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SELF-ORGANIZED ADAPTIVE PATHS IN MULTI-ROBOT MANUFACTURING:

RECONFIGURABLE AND PATTERN-INDEPENDENT
FIBRE DEPLOYMENT

CATRIONA ESCHKE, MARY KATHERINE HEINRICH,
MOSTAFA WAHBY, AND HEIKO HAMANN

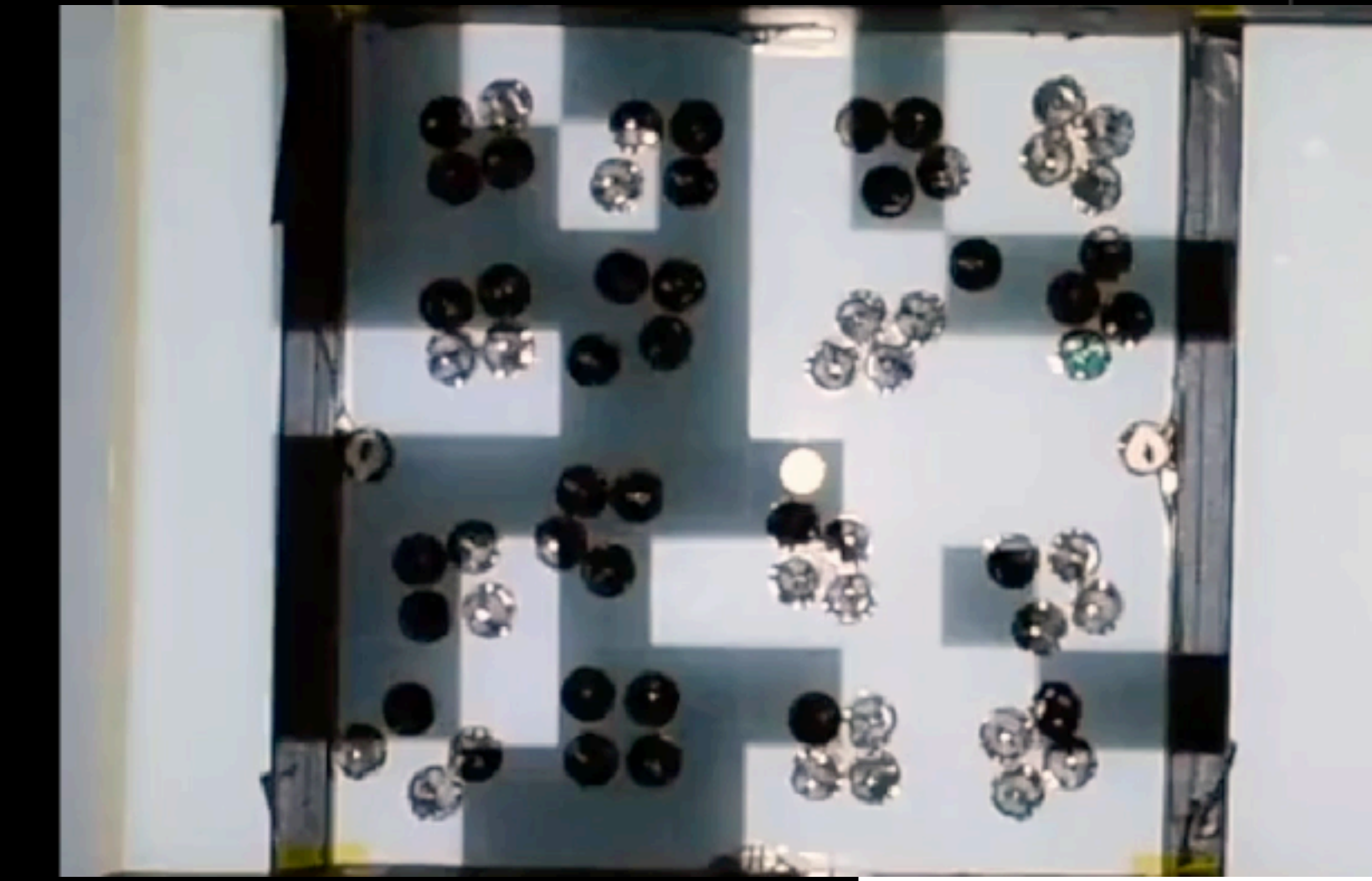


Supplementary video submission to IEEE ICRA 2023

Estimation of continuous environments by robot swarms: Correlated networks and decision-making

Mohsen Raoufi, Pawel Romanczuk, Heiko Hamann





Heiko Hamann

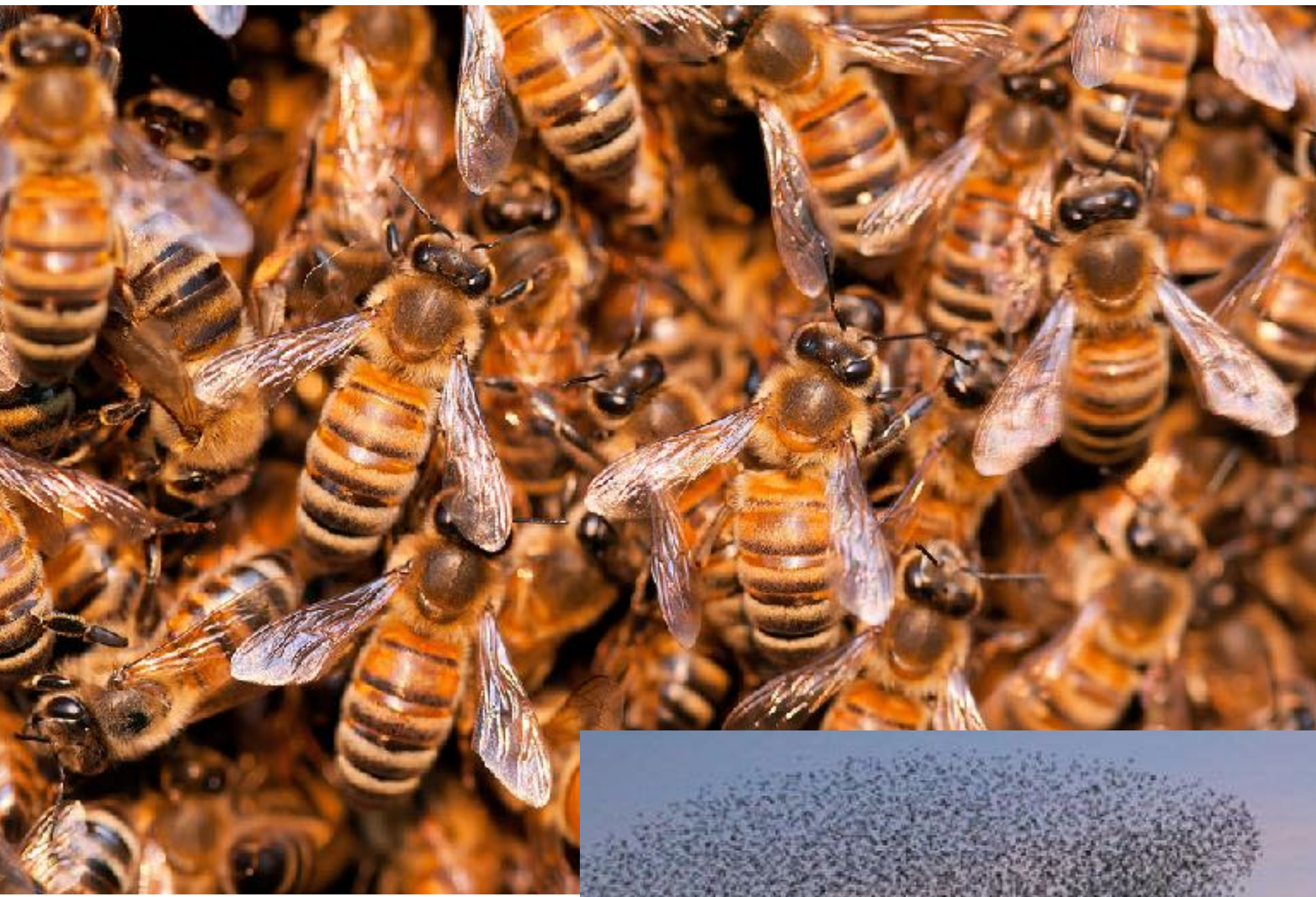
Swarm Robotics

A Formal Approach

 Springer

- 2nd edition appeared in June 2026

Canonical examples



Swarm Robotics

→ a design and synthesis perspective
(and insights from social insects)

AI “swarms” meet swarm intelligence

the questions

- What do **AI agents** do as a **collective?** (agent-agent dynamics)
- What do they do to **human collectives?** (agent-human dynamics)

What swarm intelligence offers

- Swarm intelligence/robotics offers work on **collective decision-making**, as well as models for **how minorities steer collectives**.
- *Understanding:* Are LLM collectives **swarms?**
- *Methodologies:* micro-macro models, validated on **robot swarms**
- *Interventions:* **mechanisms** that make collectives **robust**

Swarm robotics is...

- a rather **mechanistic** view
- **utilitarian**/engineering: we are done once it works
- inherent **motivation** to conform and **cost** of being wrong in biological systems:
danger of splitting (conformity),
but also danger of sub-optimal nest selection (inaccuracy)
- also consider **fission-fusion** processes (dynamical processes in which groups repeatedly split and merge, resulting in time-varying group sizes, compositions, and numbers of groups)

Defining features of a swarm



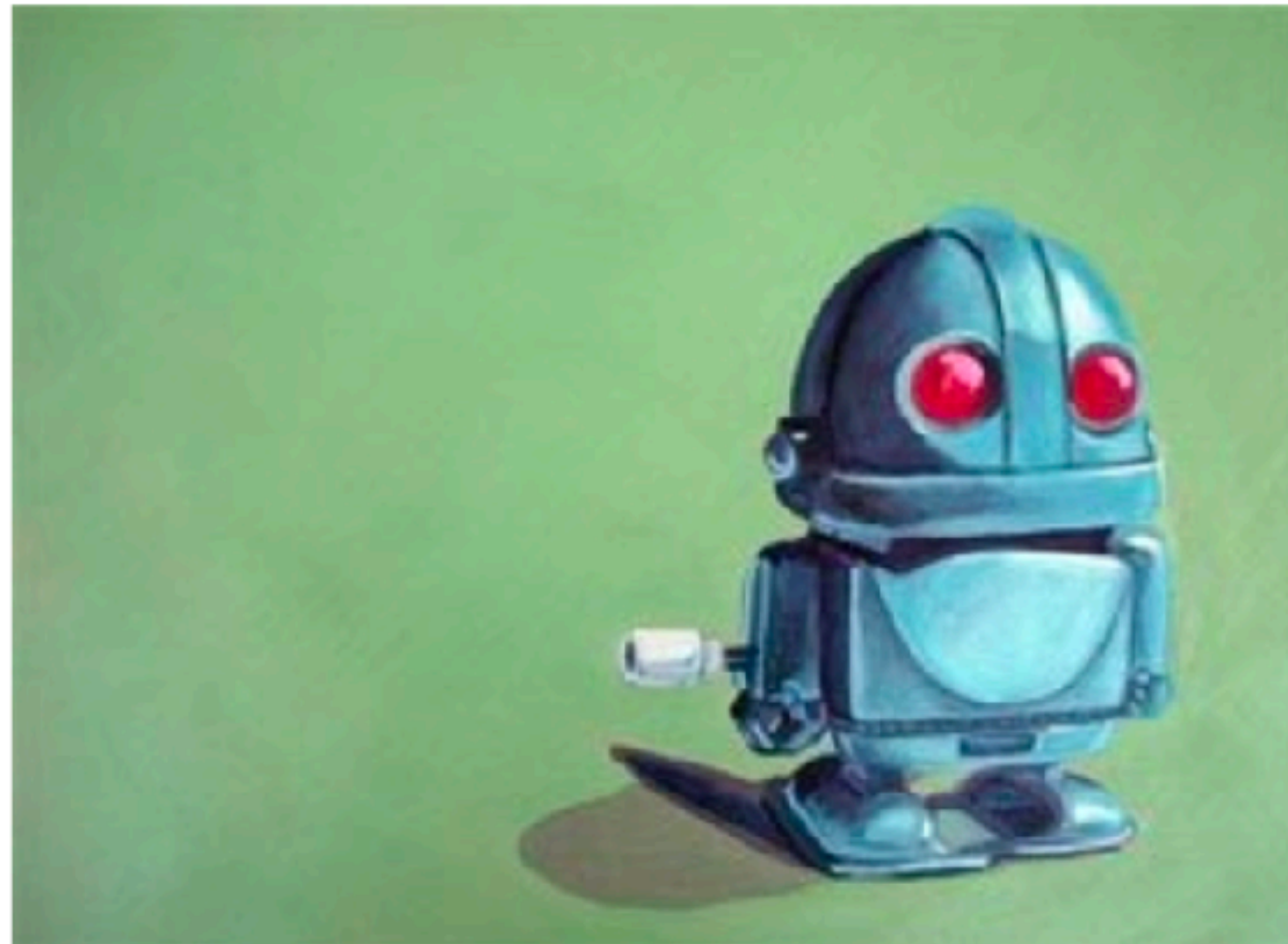
Swarm Robotics (Dorigo and Sahin, 2004)

Swarm robotics is the study of how a **large number** of relatively **simple, physically embodied** agents can be designed such that a desired collective behavior emerges from the **local interactions** among agents and between the agents and the **environment**.

decentralized
local communication / gossiping
simple reactive behaviors

complex capabilities through collaboration

The micro-macro link/problem



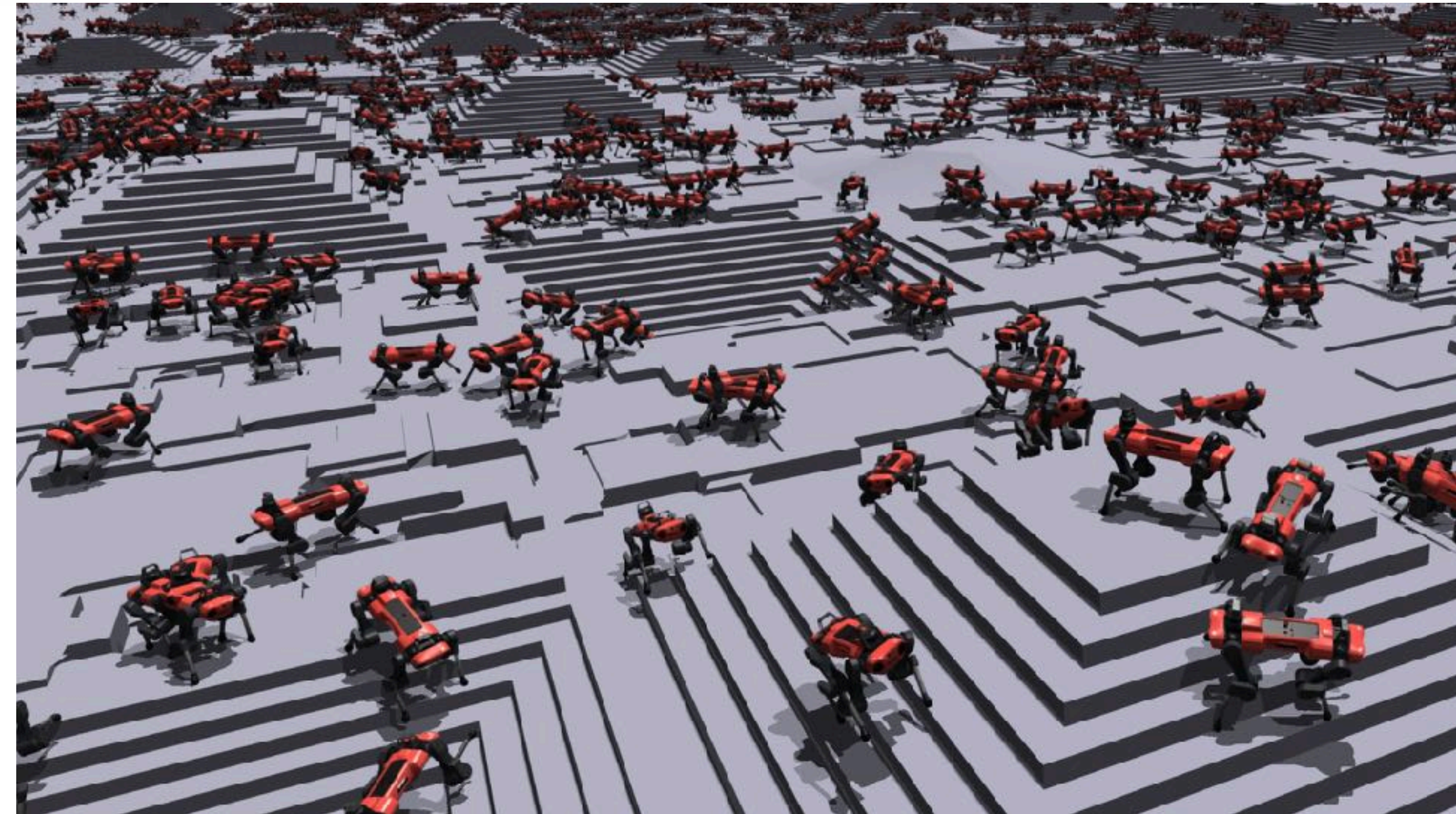
2 directions

- **analysis:** local rules → predicted global behavior
- **synthesis:** desired global behavior → local rules
- a grand challenge of swarm robotics



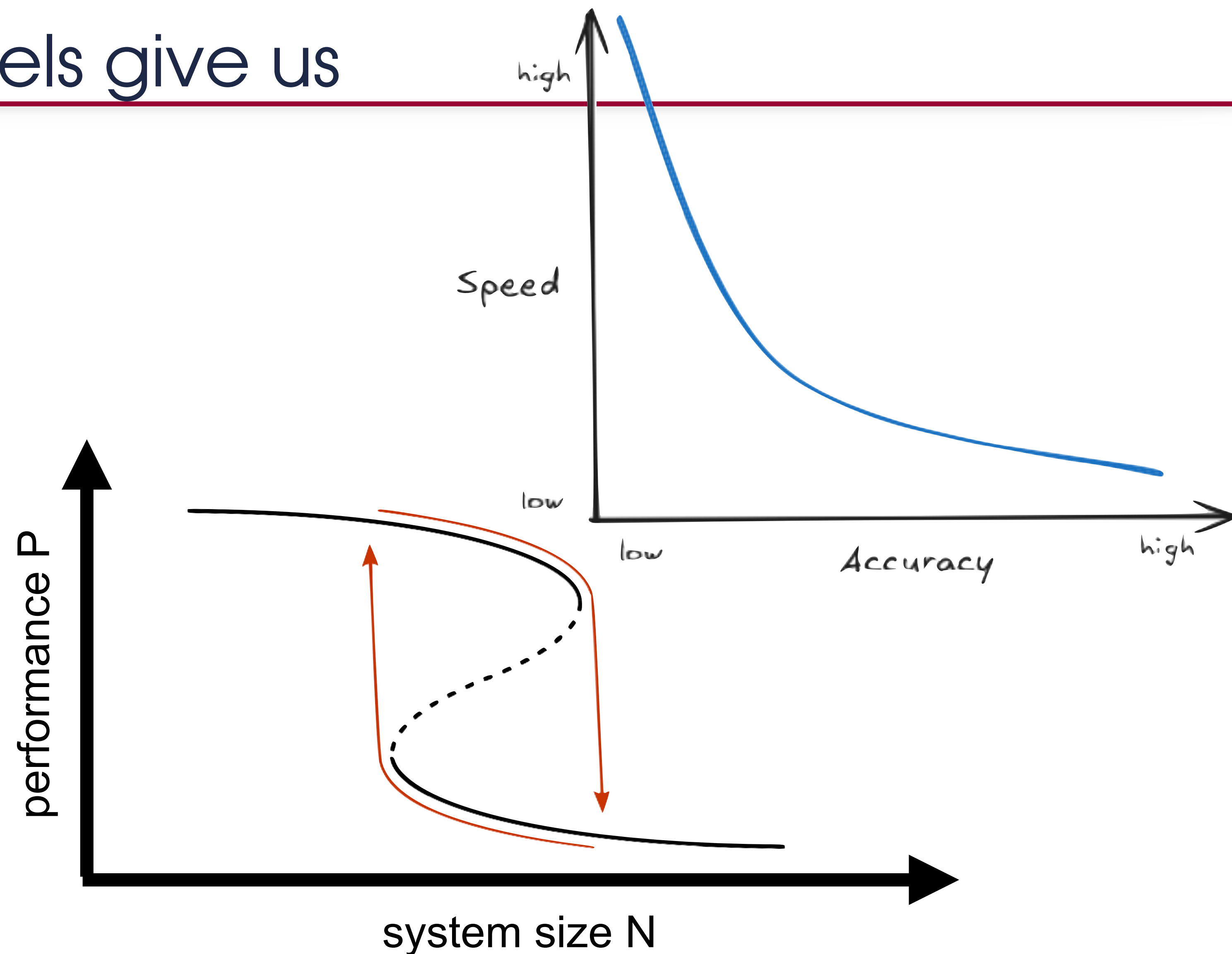
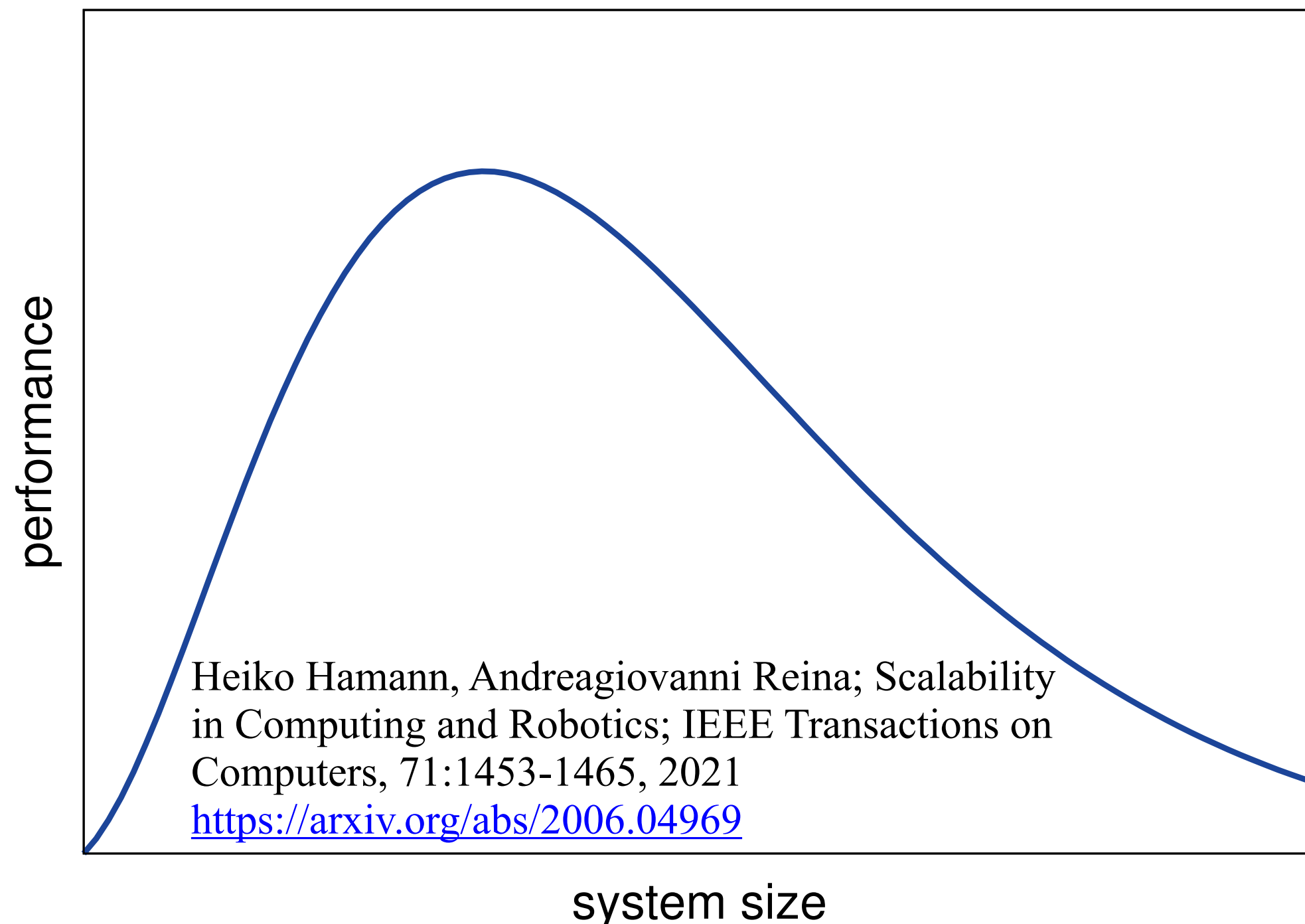
Modeling toolbox

- (robot) simulations!
 - rate equations
 - master equations
 - urn models
 - mean-field
 - Fokker–Planck approximations
-
- **assumptions:** well-mixed population, large N , Markovian agents



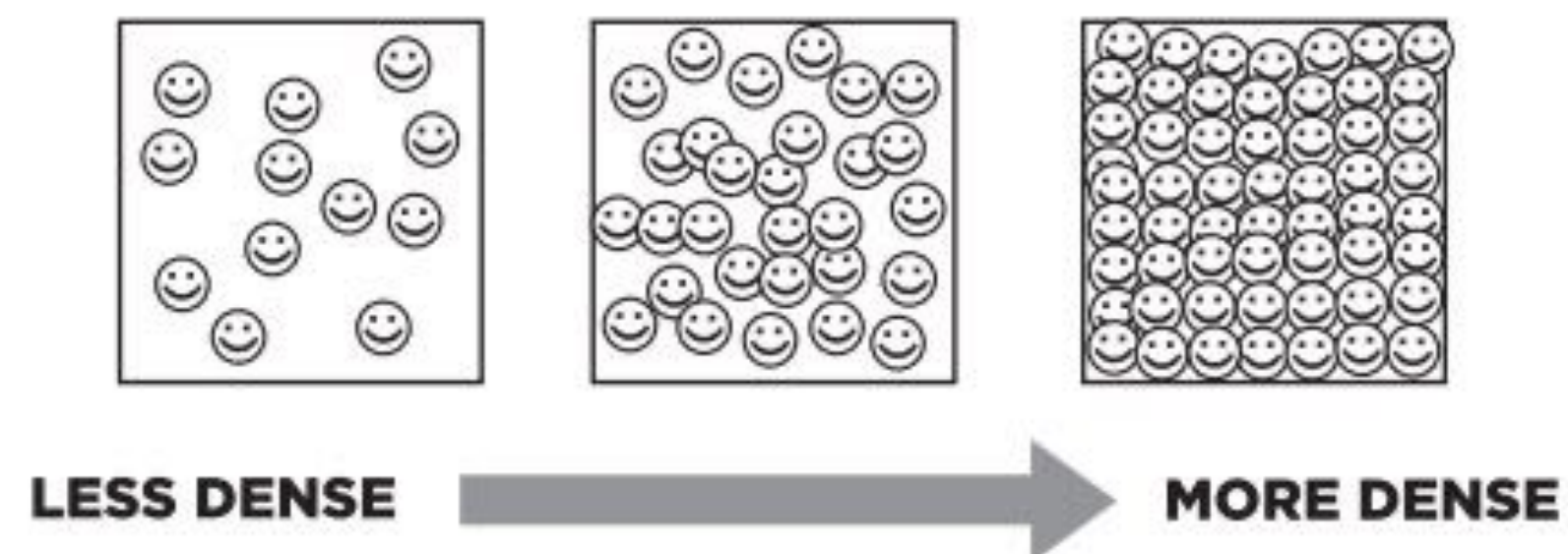
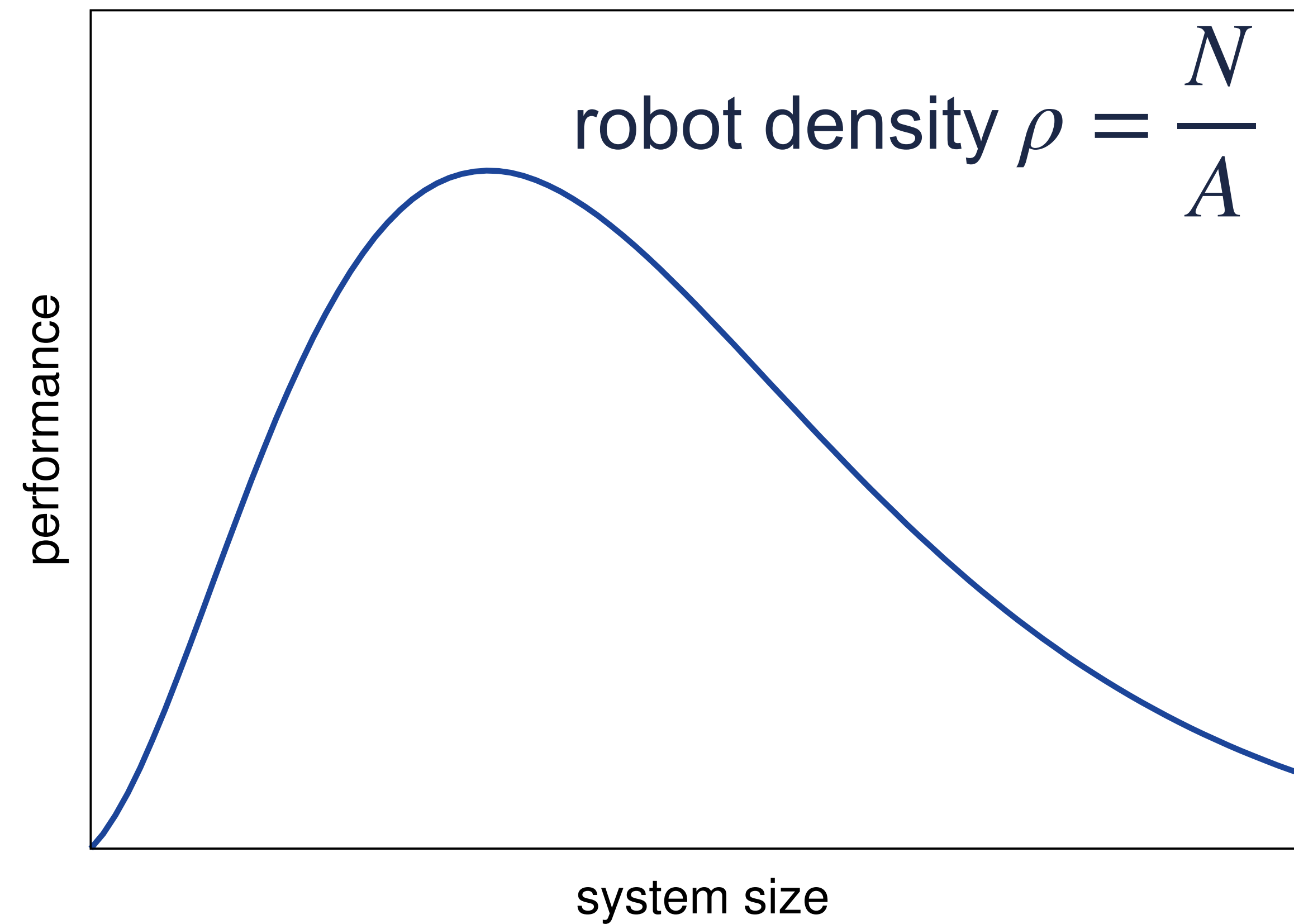
What these models give us

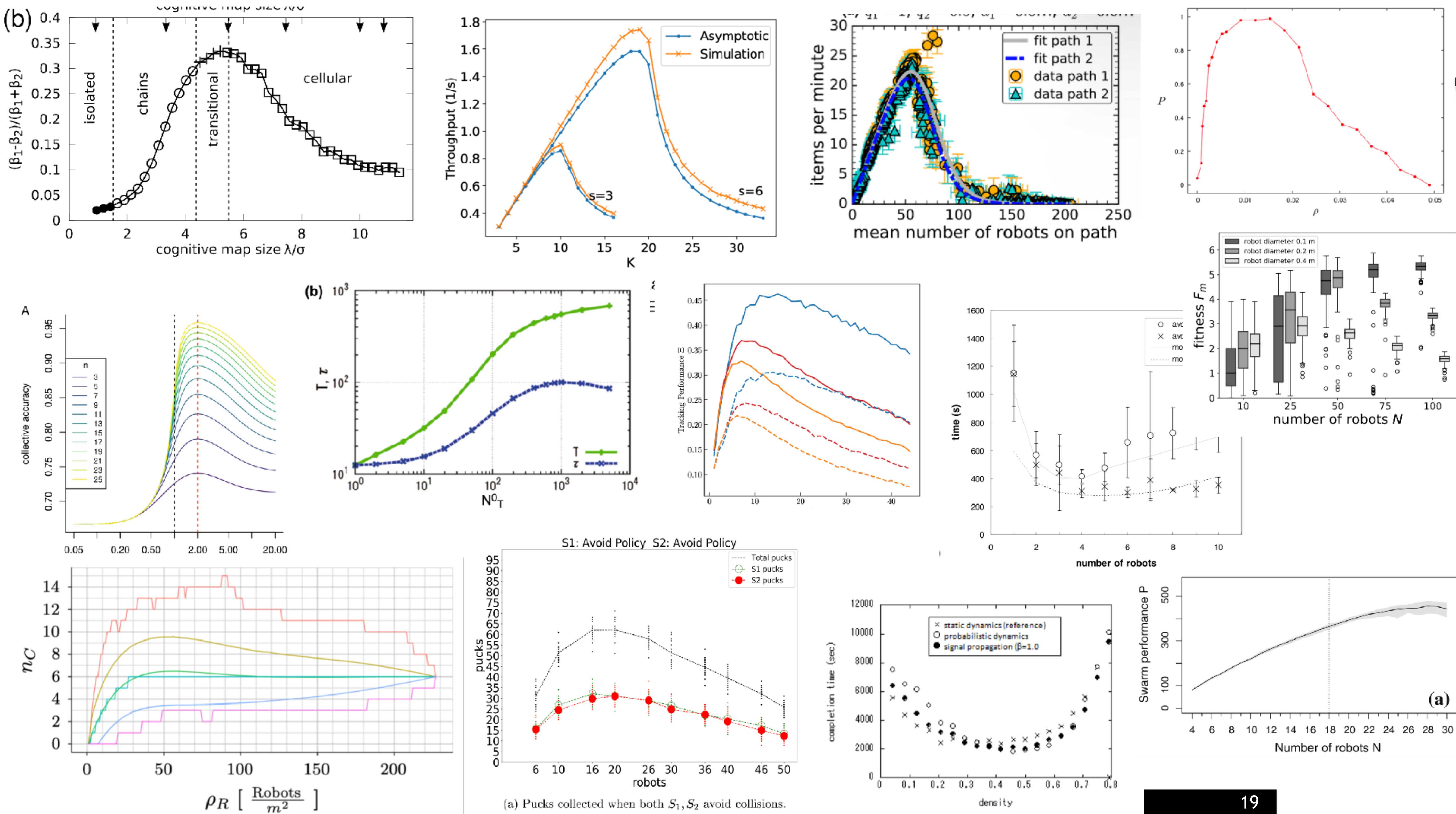
- phase transitions
- tipping points
- scaling laws
- deadlocks
- speed–accuracy tradeoffs



Jonas Kuckling, Robin Luckey, Viktor Avrutin, Andrew Vardy, Andreagiovanni Reina, Heiko Hamann; Do We Run Large-scale Multi-robot Systems on the Edge? More Evidence for Two-phase Performance in System Size Scaling; ICRA, IEEE, 2024 *IEEE ICRA Best Paper Award on Multi-Robot Systems*
<https://arxiv.org/abs/2310.11843>

system size scalability





synchronization success over connectivity

Bimodal Synchronization Performance: Why Noise and Sparse Connectivity Can Improve Collective Timing

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Andreagiovanni Reina^{1,2,3}[0000-0001-
Hamann^{1,2}[0000-

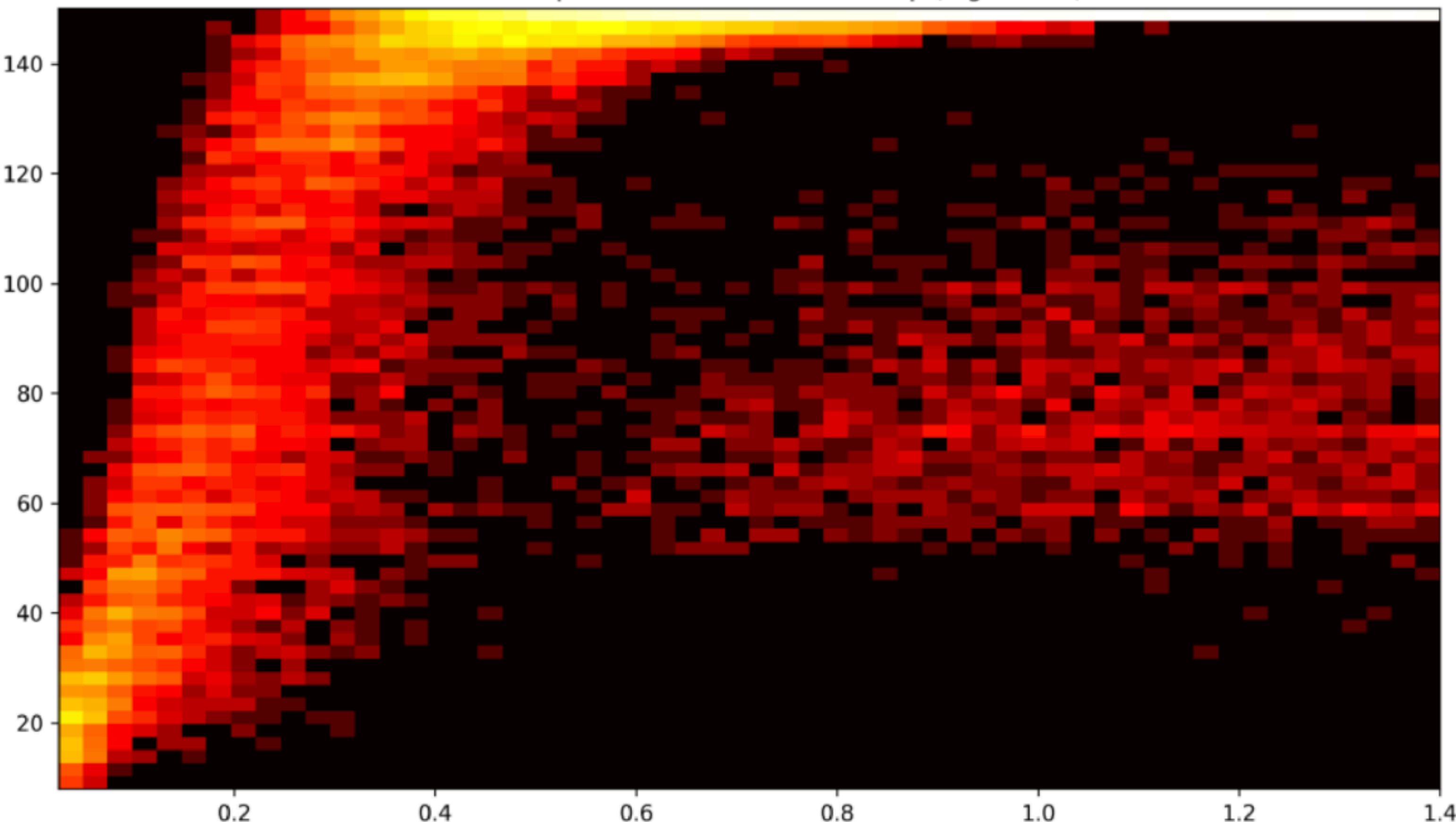
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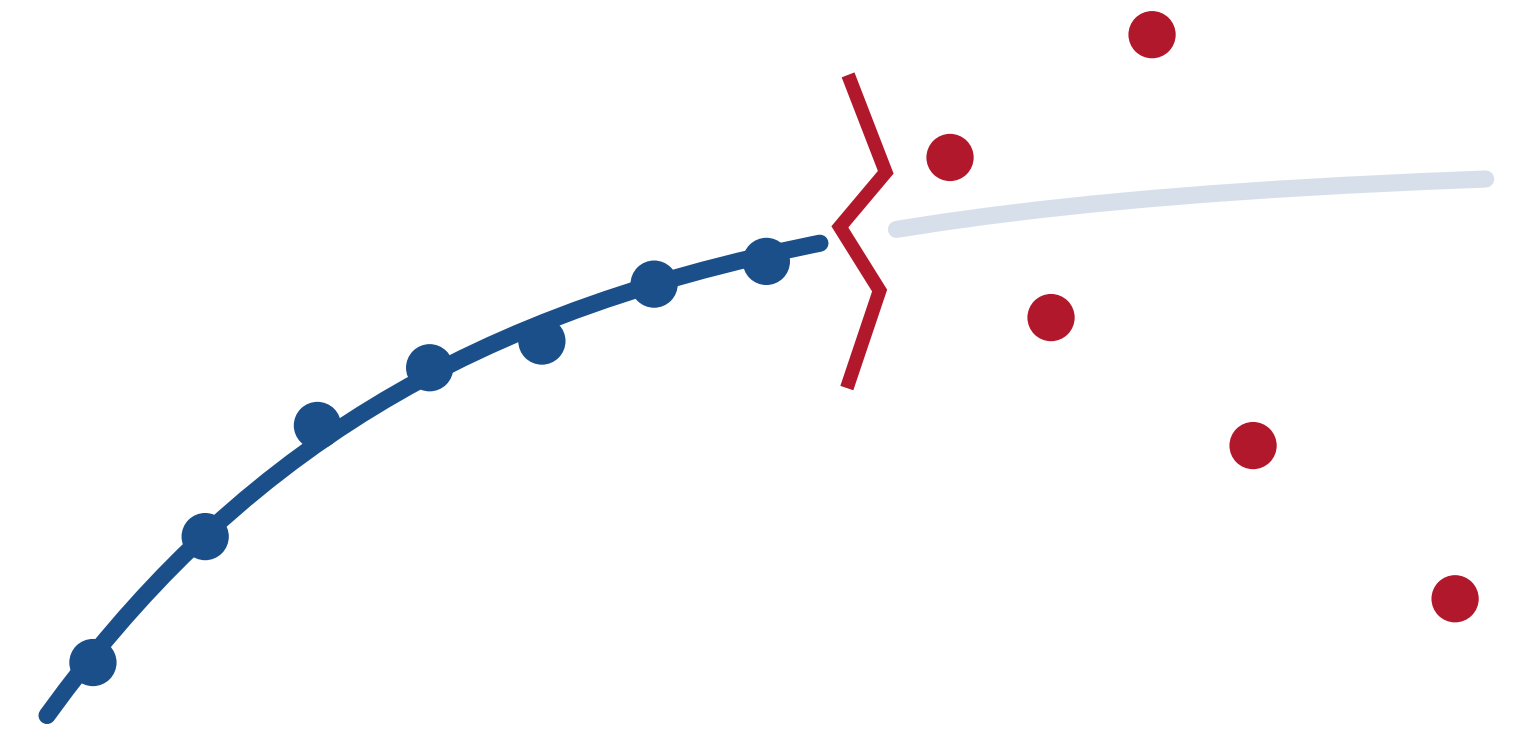
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Abstract. Pulse-coupled oscillator
nization are widely used to study
distributed systems. We analyze a c
inspired synchronization model and s
ges only near a critical balance bet



Where the models break

- small N
- heterogeneity
- agents with memory (i.e., non-Markovian)
- spatial correlations (i.e., not well-mixed)
- higher-order interactions (not only pairwise)
- agents have priors and biases (instead of fixed, simple rules)



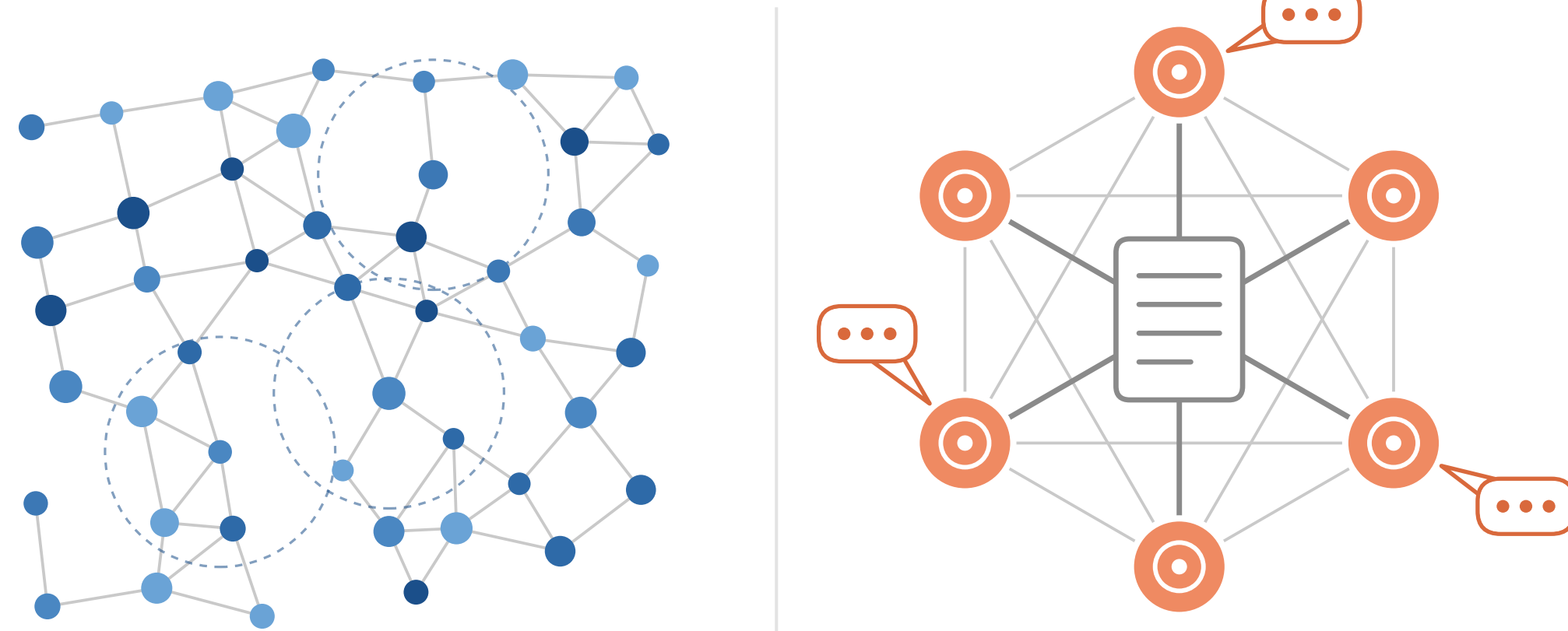
Biancalani, T., Dyson, L. & McKane, A. J. Noise-induced bistable states and their mean switching time in foraging colonies. Phys. Rev. Lett. 112, 038101 (2014).

Khaluf, Y., Pinciroli, C., Valentini, G. & Hamann, H. The impact of agent density on scalability in collective systems: noise-induced versus majority-based bistability. Swarm Intell. 11, 155–179 (2017).

Stark, H.-U., Tessone, C. J. & Schweitzer, F. Decelerating microdynamics can accelerate macrodynamics in the voter model. Phys. Rev. Lett. 101, 018701 (2008).

Battiston, F. et al. Networks beyond pairwise interactions: Structure and dynamics. Phys. Rep. 874, 1–92 (2020).

Classical swarm vs. AI collective



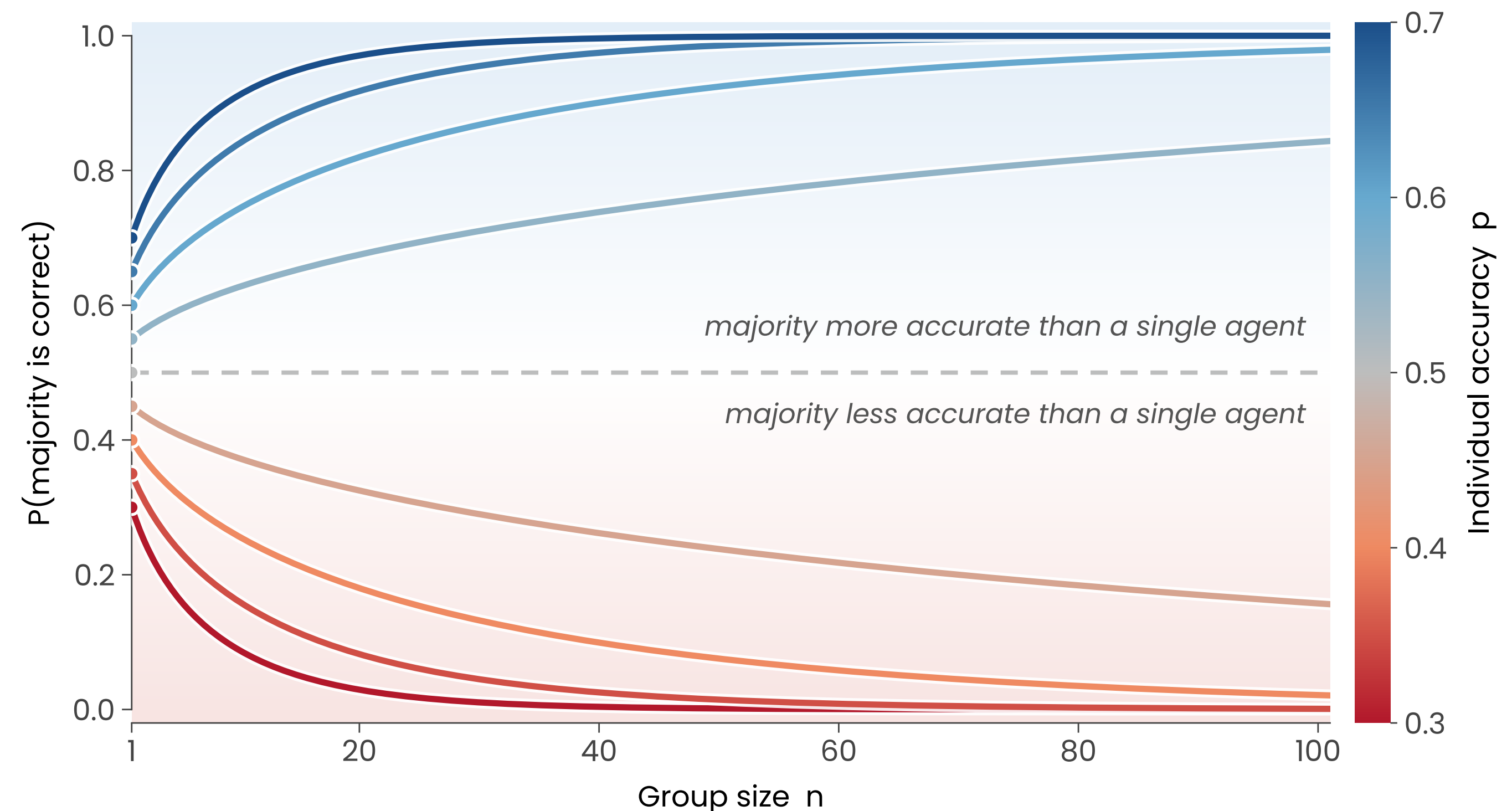
- agent **cognition**:
reactive vs. deliberative
- **information scope**:
local vs. shared/global
- channel: **signals vs. language**
- **topology**:
spatial/local vs. fully connected or hub-based
- **diversity**:
individual variation
vs. copies of the same model
- **goals**:
all share the same goal
vs. agents choose/switch goals online

Correlated errors vs diversity

- Wisdom of the crowds and Condorcet theorem require **independence and diversity**
- identical model copies with shared context are not diverse/independent



ChatGPT Classic Version 1.2026.184 (1784145287)



Can SI models predict AI collectives?

Yes, maybe

- universality: macro dynamics depend on a few features
- coarse-graining: answers are discrete states

No, not directly

- small N ? correlated copies: the effective N is maybe smaller
- memory: agents know a lot and may even communicate globally
- content matters, not just counts

Observed collective failure modes

AI failure mode	Swarm counterpart	Swarm remedy
premature consensus	stagnation or premature convergence in ant colony and particle swarm optimization	pheromone evaporation, bounds on trail strength
information cascades	ant mill: ants following each other's trail in a circle until they die	negative feedback, independent sampling
sycophantic convergence	copying without evaluating quality, as in the voter model	quality-weighted advocacy (weighted voter model)
orchestrator error amplification	centralization, a single point of failure	decentralization, redundancy
debates collapsing into agreement	recruitment without stop signals	cross-inhibition

Would more decentralization help?

pro

- preserved diversity
- fewer cascades
- no single point of failure
- bounded context per agent

contra

- slower convergence
- coordination overhead
- wasted capability



Philosophy of Science

Article contents

Abstract
Footnotes
References

The Communication Structure of Epistemic Communities

Published online by Cambridge University Press: 01 January 2022

[Kevin J. S. Zollman](#)

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Abstract

Increasingly, epistemologists are becoming interested in social structures and their effect on epistemic enterprises, but little attention has been paid to the proper distribution of experimental results among scientists. This paper will analyze a model first suggested by two economists, which nicely captures one type of learning situation faced by scientists. The results of a computer simulation study of this model provide two interesting conclusions. First, in some contexts, a community of scientists is, as a whole, more reliable when its members are less aware of their colleagues' experimental results. Second, there is a robust tradeoff between the reliability of a community and the speed with which it reaches a correct conclusion.

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RESEARCH ARTICLE | COLLECTIVE BEHAVIOR

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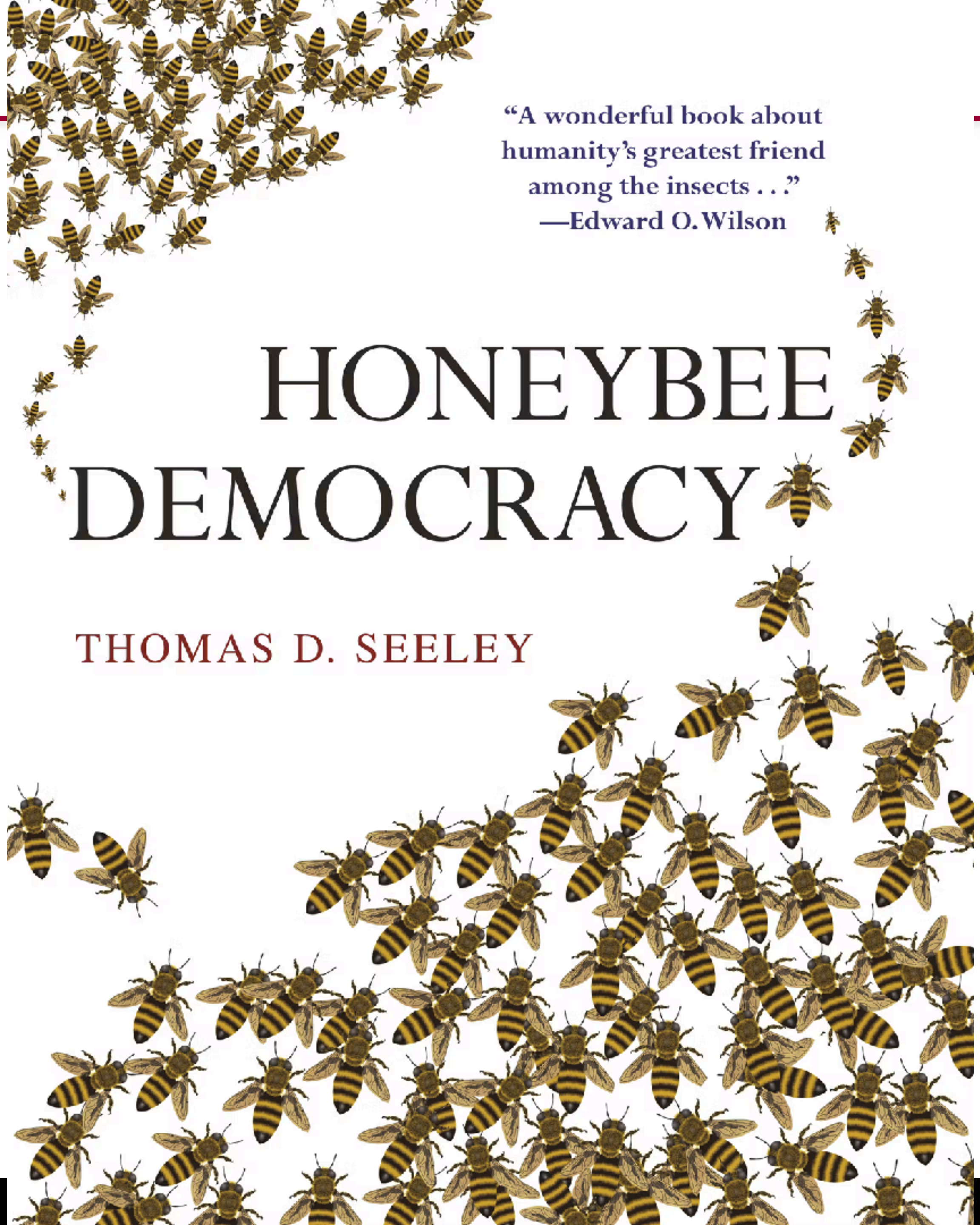
When less is more: Robot swarms adapt better to changes with constrained communication

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“A wonderful book about
humanity’s greatest friend
among the insects . . .”
—Edward O. Wilson

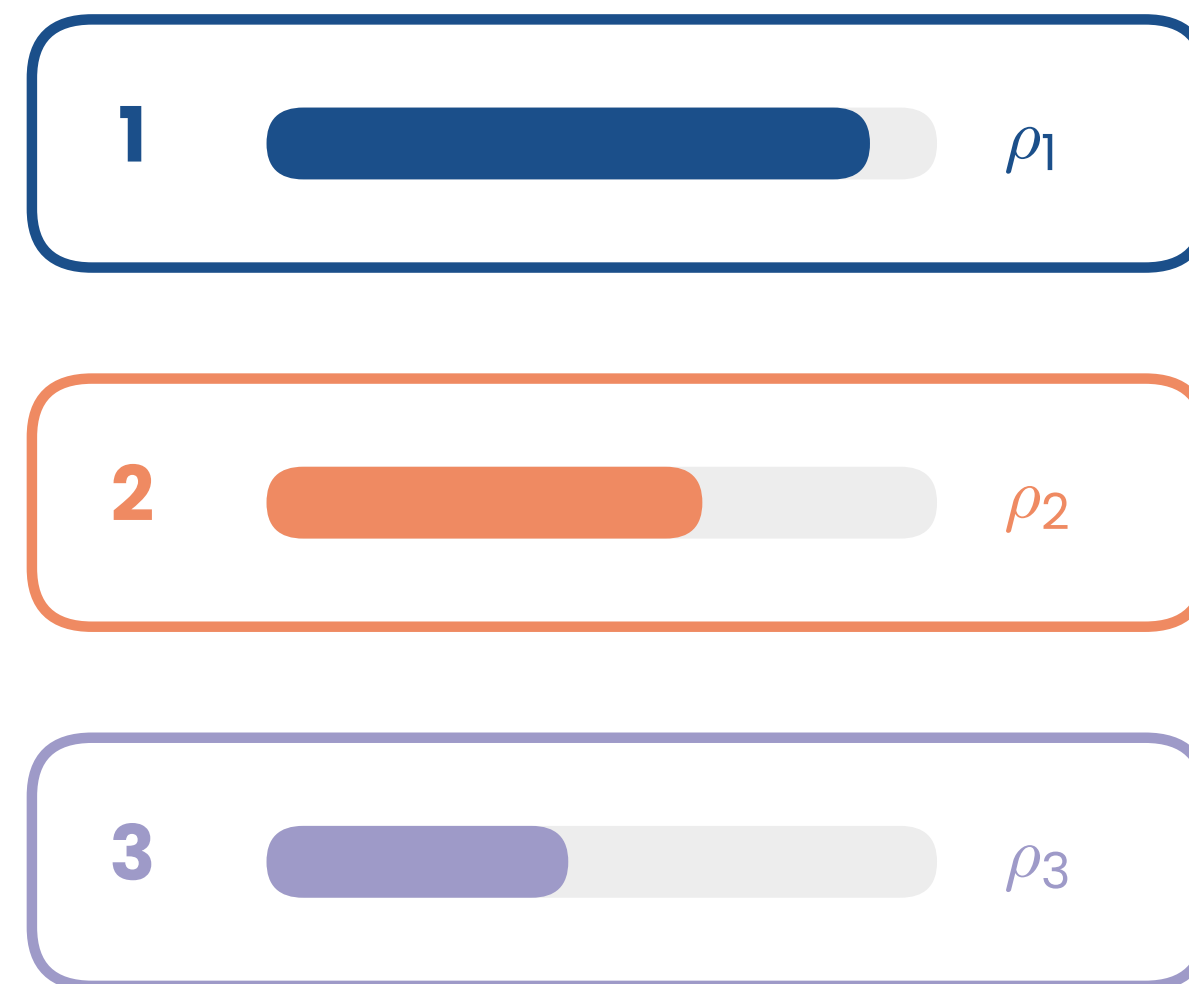
HONEYBEE DEMOCRACY

THOMAS D. SEELEY

The best-of-n problem

n options

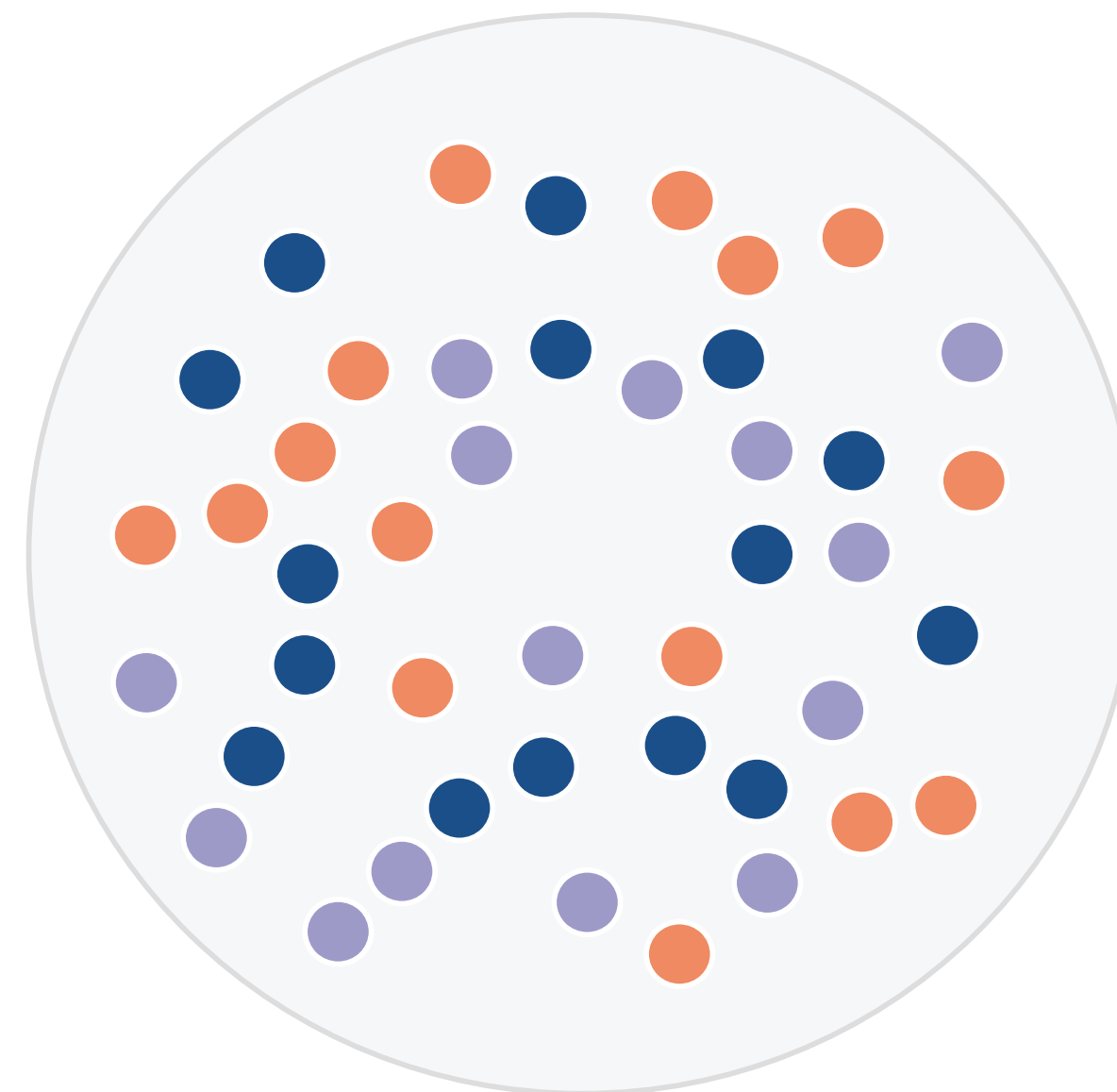
qualities unknown to the agents



$$\rho_1 > \rho_2 \geq \dots \geq \rho_n$$

N agents

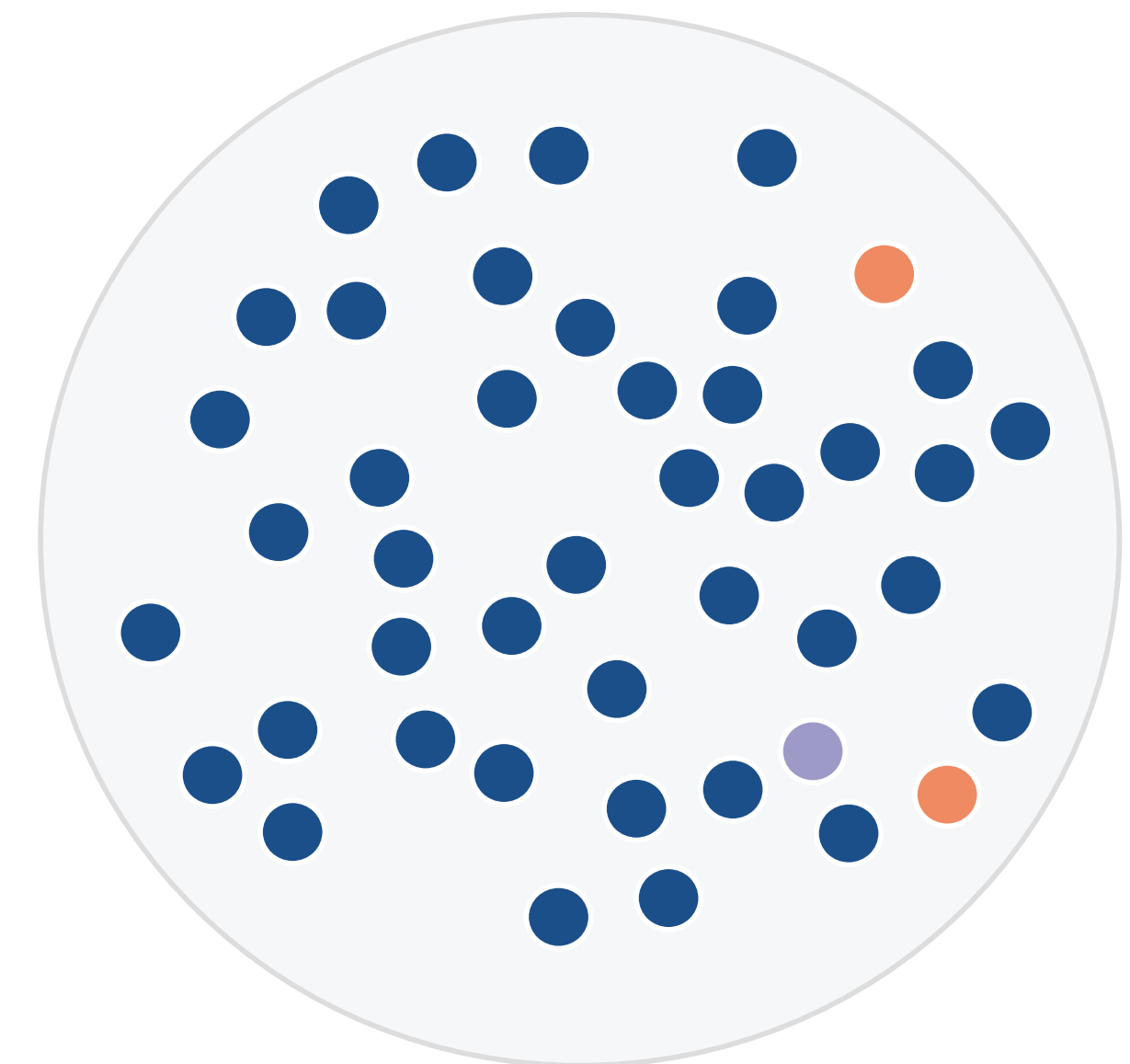
each holds an opinion $o_j \in \{1, \dots, n\}$



local interactions, noisy quality estimates

Consensus

on the best option



$$x_1 \geq 1 - \delta$$

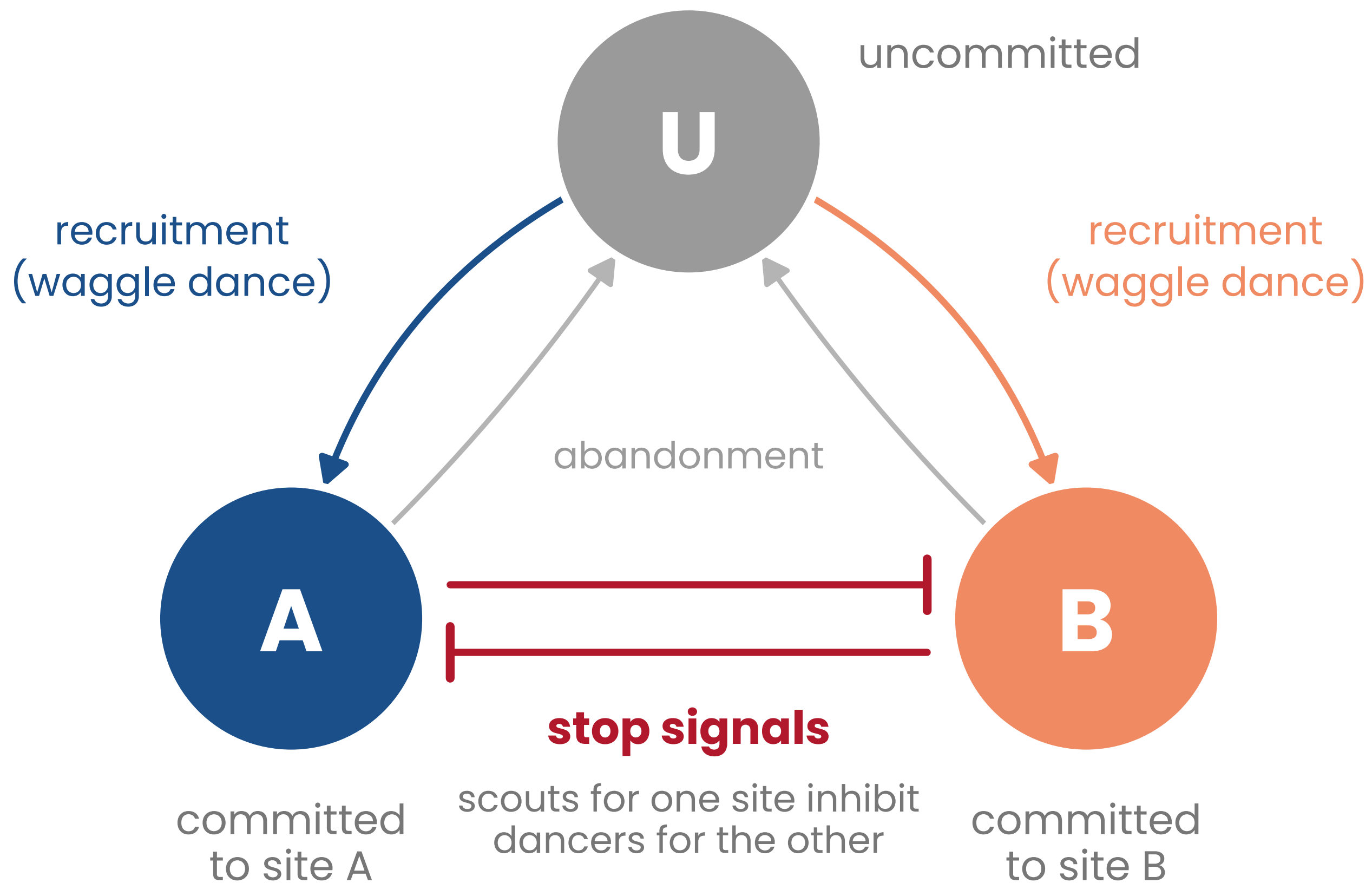
x_1 : fraction of agents with opinion 1

Goal: reach consensus on the best option as fast as possible (speed) and as reliably as possible (accuracy)

Honeybees and stop signals

Mechanism

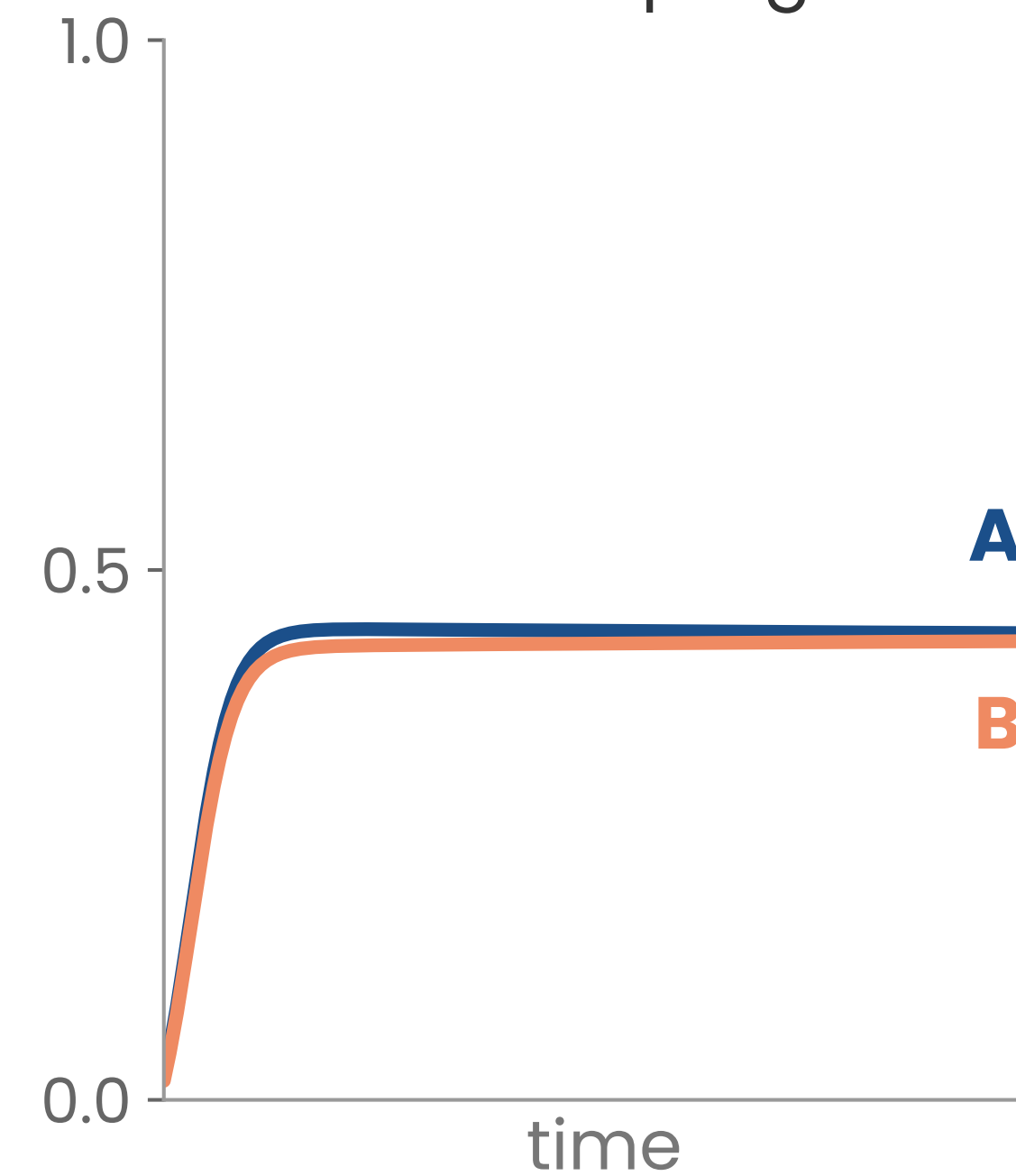
two nest sites of equal quality



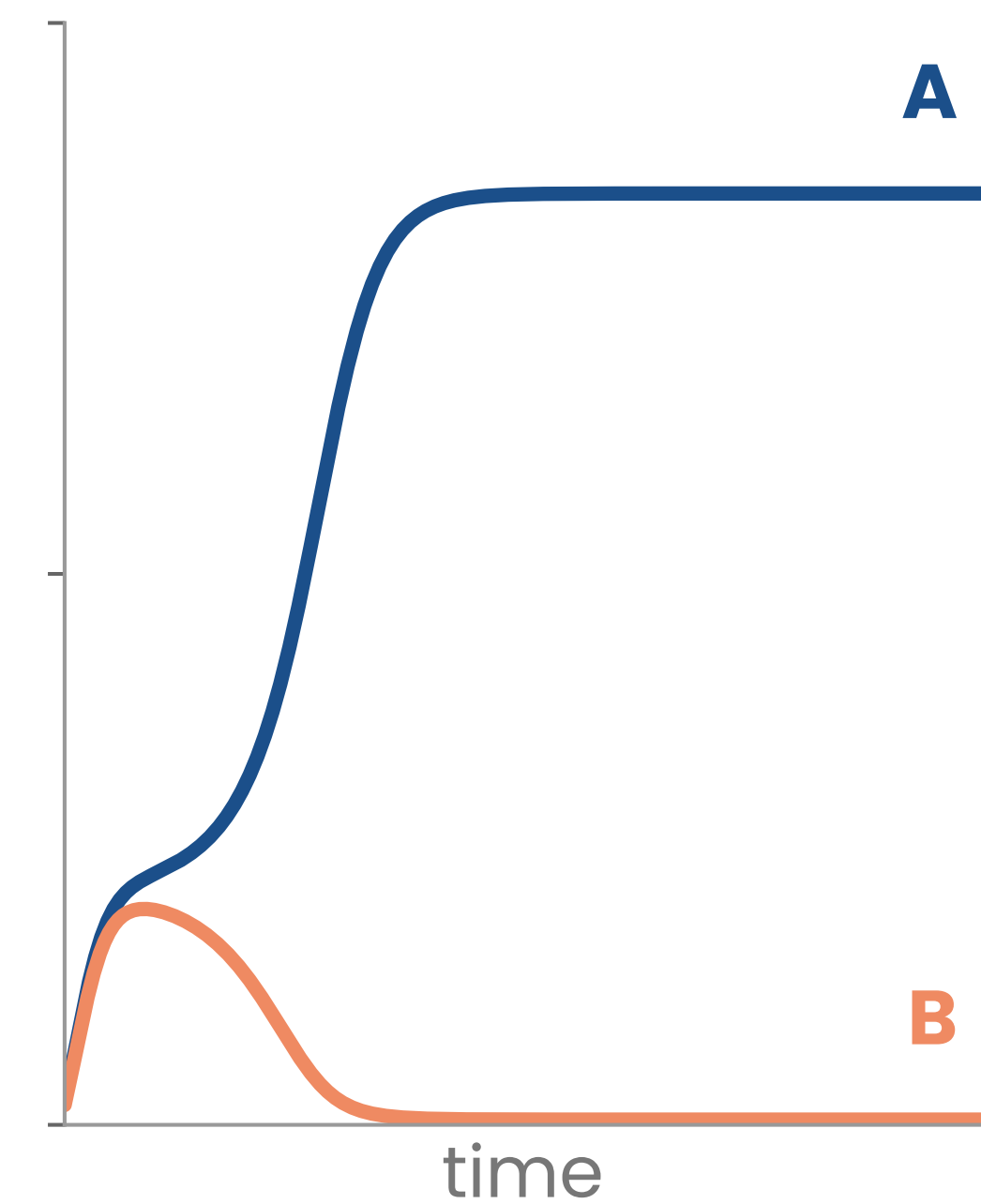
Outcome

fraction of scouts committed to each site

without stop signals



with stop signals



after Seeley et al., Science 2012



The cross-inhibition model

population fractions committed to options A, B

$$\dot{X}_A = \underbrace{\gamma X_U}_{\text{commitment}} - \underbrace{\alpha X_A}_{\text{abandonment}} + \underbrace{\rho X_U X_A}_{\text{recruitment}} - \underbrace{\sigma X_A X_B}_{\text{cross-inhibition}}$$

$$\dot{X}_B \text{ analogous, } X_U = 1 - X_A - X_B$$

equal options of value v : $\gamma = \rho = v$, $\alpha = 1/v$ $v > 1$

$$\text{deadlock breaks at } \sigma^* = \frac{4v^3}{(v^2 - 1)^2}$$

higher value $\rightarrow$ smaller stop signal needed

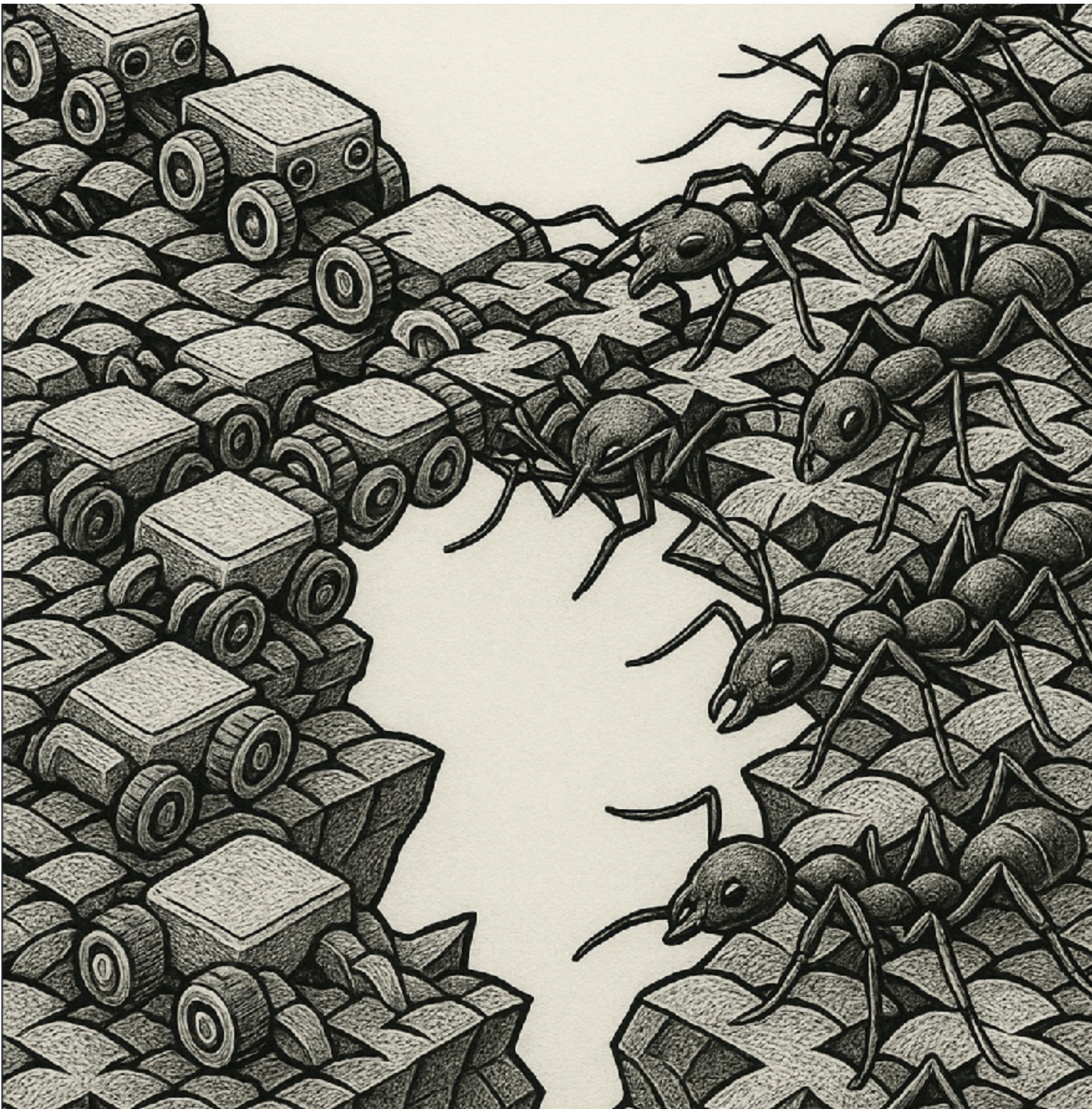
Mapping to AI agents

Term	Honeybees	LLM agents
commitment γ	scout discovers a site	agent forms an answer from its own evidence
recruitment ρ	waggle dance	agent argues for its answer to undecided agents
abandonment α	scout stops dancing	agent's confidence decays, it rechecks
cross-inhibition σ	stop signal to rival dancer	targeted counter-evidence against the rival answer

Other mechanisms worth porting?

Mechanism	Origin / applied in	AI analogue	Reference (example)
Quorum threshold	honeybees	stop at quorum, not unanimity	Seeley & Visscher, Behav. Ecol. Sociobiol. 2004
Majority rule	robot swarms	poll several peers, not one	Valentini et al., JAAMAS 2016
Weighted voting	robot swarms	argue longer for better answers	Valentini, Hamann & Dorigo, AAMAS 2014
Informed minority	fish schools	few agents hold key evidence	Couzin et al., Nature 2005
Bounded confidence	opinion dynamics	listen only to nearby views	Hegselmann & Krause, JASSS 2002
Stigmergy	e.g., ant trails	shared artifact, not chat	Theraulaz & Bonabeau, Artif. Life 1999

Questions



- Which would you trust more: one strong agent, or a crowd of agents?
- Should AI collectives be **designed to disagree**?
- What is the **stop signal for an LLM agent**?